

Geometric Stability of Neural Population Codes: Regional Variation, Behavioral Relevance, and Circuit Dependence

Prashant C. Raju

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Abstract

Current models of representational reliability in neural populations focus on temporal stability: whether population centroids are preserved across sessions and days. This framing leaves a fundamental question unanswered: how reliably does the pairwise distance structure among stimuli reproduce across independent observations within a session? We argue that this property, geometric stability, constitutes an independent axis of representational analysis that existing frameworks do not capture. We formalize geometric stability as the Spearman rank correlation between split-half representational

dissimilarity matrices (Shesha) and show that it is empirically dissociable from both temporal stability and decoding accuracy. Across 229 area-session observations spanning 68 brain regions in a visual discrimination task (Steinmetz et al. 2019), geometric stability predicts trial-by-trial neural-behavioral coupling ($\rho = 0.18$, $p = 0.005$) while centroid drift does not ($\rho = 0.002$, $p = 0.976$). The regional hierarchy, with striatum most stable ($\bar{S} = 0.44$) and hippocampus least ($\bar{S} = 0.19$), runs roughly opposite to the temporal stability hierarchy. Directionally consistent olfactory data (Bolding & Franks 2018) motivate an attractor network model in which recurrent excitatory coupling amplifies split-half RDM consistency by completing stimulus patterns from sparse feedforward input ($\rho = +0.64$, $p = 0.010$), providing a circuit-level account of how geometric stability emerges. These results establish geometric stability as a functionally relevant, circuit-dependent property of neural population codes, orthogonal to temporal drift measures and complementary to recent accounts of how recurrent connectivity balances representational stability with sequential dynamics in hippocampal circuits.

1 Introduction

Neural populations are noisy. The same stimulus, presented twice to the same animal, elicits different responses: different cells, different rates, different timing. Yet animals identify odors reliably, discriminate contrasts, execute learned movements. The variability is real; so is the stability. Something is not varying. The dominant answer has been centroid preservation: population-averaged responses to each condition maintain their positions in neural state space even as individual neurons reorganize (Gallego

et al., 2020; Rule et al., 2019). But a population’s utility to downstream areas depends not only on where centroids sit but on whether the full relational structure among conditions is consistent across observations. How recurrent circuitry balances the preservation of stable representations with the capacity for flexible, temporally structured dynamics is a central open question across brain regions (Wagner et al., 2026; Morales et al., 2025). We focus on one side of this balance: the reliability of representational geometry within a session.

The dominant explanation is that individual neuron contributions fluctuate while low-dimensional latent dynamics are preserved (Gallego et al., 2020). Representational drift (Driscoll et al., 2017; Ziv et al., 2013; Rule et al., 2019)—the gradual shift in which cells encode what—has been documented across hippocampus, motor cortex, and piriform cortex. The framing is temporal: does the population centroid stay put over sessions and days?

That framing misses something. Centroid preservation tells us the average population state is maintained. It does not tell us whether the *relational structure* among stimuli is preserved: whether the pattern of pairwise distances between conditions is consistent from one set of trials to the next. Representational similarity analysis (RSA) (Kriegeskorte et al., 2008) and its extensions (Nili et al., 2014; Walther et al., 2016; Diedrichsen & Kriegeskorte, 2017) characterize pairwise geometry, but neither measures how *reliably* that geometry appears across independent observations. Reliability, not just content, is what we ask about here.

One might ask whether decoding accuracy already captures this. It does not. Decoding asks whether task variables can be read out from a population: a question about

information content. Geometric stability asks whether the pairwise distance structure among conditions reproduces reliably across independent trial subsets: a question about representational architecture. The two are empirically orthogonal in our data ($\rho = 0.09$, $p = 0.19$, $n = 228$): areas with high decoding accuracy are not systematically more or less geometrically stable. Yet geometric stability predicts trial-by-trial neural-behavioral coupling ($\rho = 0.18$, $p = 0.005$) while decoding accuracy does not ($\rho = 0.01$, $p = 0.88$). A population can be highly decodable yet geometrically brittle. If information is concentrated in a low-dimensional subspace, linear readout succeeds while split-half RDM consistency fails. Shesha is sensitive to this failure mode; decoding is not.

We quantify geometric stability using Shesha, a measure we introduced recently (Raju, 2026a,b). Shesha is a general-purpose metric for representational reliability, applicable wherever observations can be split into independent subsets: condition-averaged RDMs (Kriegeskorte et al., 2008) are computed on each half, and geometric stability is their Spearman rank correlation. It has been validated across artificial neural networks, protein sequences, and single-cell molecular profiles (Raju, 2026a,c,d); here we apply it to neural electrophysiology for the first time in a substantive neuroscience context. Shesha is empirically orthogonal to representational similarity metrics: across 2,463 encoder configurations spanning seven domains, geometric stability and centered kernel alignment share less than 0.1% of variance ($\rho = -0.01$), and Shesha detects compression artifacts from aggressive dimensionality reduction that similarity measures miss entirely (Raju, 2026a). The primary aim of this paper is to go beyond validation and ask substantive neuroscience questions: whether within-session geometric reliability

varies systematically across brain regions, whether that variation predicts behavior, and whether it depends on circuit architecture in the way an attractor account would predict.

Trials are split into odd and even subsets. Condition-averaged population vectors are computed on each half, and pairwise cosine distances give two RDMs:

$$D_{ij}^{(s)} = 1 - \frac{\bar{\mathbf{x}}_i^{(s)} \cdot \bar{\mathbf{x}}_j^{(s)}}{\|\bar{\mathbf{x}}_i^{(s)}\| \|\bar{\mathbf{x}}_j^{(s)}\|}, \quad s \in \{1, 2\}. \quad (1)$$

Eq. 1 defines the object whose reliability we measure. Each $D^{(s)}$ captures the full pairwise geometry of conditions in one data split: it is this structure, not the centroid or any single axis, that determines whether downstream areas receive consistent relational input across trials. Geometric stability is the rank correlation between these two independent estimates:

$$\mathcal{S} \text{ (Shesha)} = \rho_s(\text{vec}(D^{(1)}), \text{vec}(D^{(2)})) . \quad (2)$$

$\mathcal{S}$ is high when the pairwise distance structure between conditions is reproducible across independent trial samples, and low when trial noise makes the RDMs inconsistent. The rank correlation in Eq. 2 is the core quantity that separates this framework from both RSA (which compares a data RDM to a model RDM) and centroid-based drift (which tracks mean population state). Shesha compares data to data, and it operates on the full relational structure rather than any summary statistic. Crucially, $\mathcal{S}$ and centroid-based drift measure different things and are empirically dissociable (Raju, 2026a), as we show below.

We apply Shesha to two electrophysiology datasets chosen because they ask complementary questions. The Steinmetz et al. (2019) Neuropixels recordings span 26 sessions and 68 brain regions during a visual discrimination task. The Bolding & Franks

(2018) PCX-1 dataset asks whether geometric stability depends on recurrent circuitry in the olfactory system.

2 Results

2.1 Striatum is more geometrically stable than hippocampus, and the ordering inverts for temporal stability

Neuropixels recordings from 26 sessions and 68 brain regions during a visual discrimination task (Steinmetz et al., 2019) gave 229 area-session observations with at least 10 simultaneously recorded neurons. For each area-session, spike counts were averaged over the 0–500 ms post-stimulus decision epoch to give one population vector per trial. Each trial vector was L2-normalized before computing RDMs. Shesha was computed from odd- and even-indexed trial splits: for each of 9 contrast-pairing conditions, condition-averaged population vectors were computed separately on each half, and Shesha is the Spearman rank correlation of the two resulting RDMs (see Section 3). Striatum was highest ($\bar{S} = 0.44$, 95% CI [0.34, 0.56], $n = 6$), followed by motor cortex (0.38, [0.25, 0.50], $n = 10$) and visual cortex (0.36, [0.27, 0.46], $n = 29$). Hippocampus was lowest (0.19, [0.13, 0.25], $n = 39$).

Centroid drift was defined as the cosine similarity between the mean L2-normalized population vectors computed separately over the first and second halves of each session (split at the median trial). The centroid for each half is the arithmetic mean of the L2-normalized trial vectors within that half. Centroid drift ran in the opposite direction to geometric stability. Thalamus drifted least (0.95, [0.92, 0.97]), hippocam-

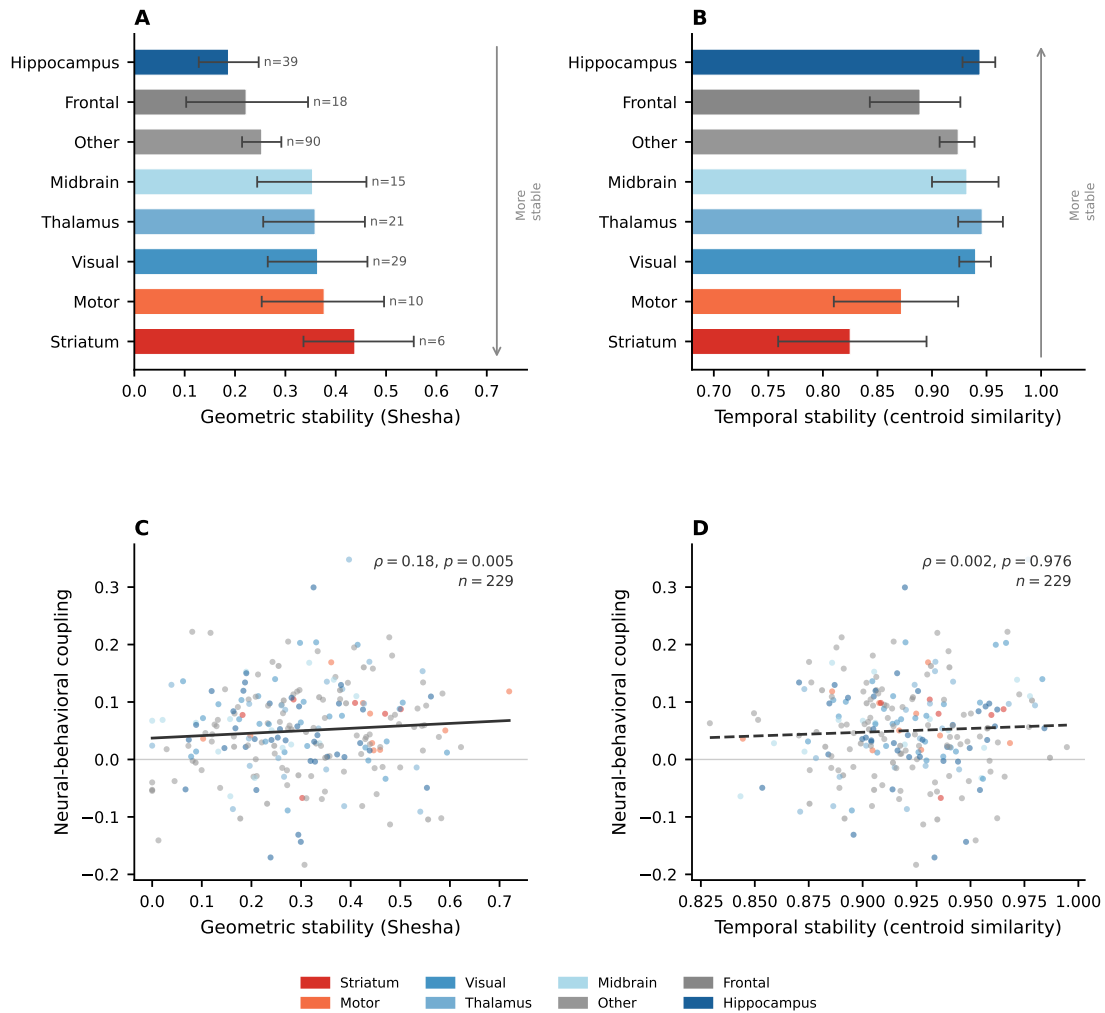


Figure 1: Geometric stability predicts neural-behavioral coupling; centroid drift does not.

Figure 1: **Figure 1 (continued): Geometric stability predicts neural-behavioral coupling; centroid drift does not.** Each point is one area-session observation ($n = 229$), colored by functional region. **(A)** Geometric stability (Shesha) vs. trial-by-trial neural-behavioral coupling (Spearman correlation between population vector magnitude and trial outcome). Solid line: linear regression. $\rho = 0.18$, $p = 0.005$. **(B)** Centroid drift vs. the same coupling measure. Dashed line: linear regression. $\rho = 0.002$, $p = 0.976$. Region color legend shown below. **(C)** Geometric stability (Shesha) vs. trial-by-trial neural-behavioral coupling (Spearman correlation between population vector magnitude and trial outcome). Solid line: linear regression. $\rho = 0.18$, $p = 0.005$. **(D)** Centroid drift vs. the same coupling measure. Dashed line: linear regression. $\rho = 0.002$, $p = 0.976$.

pus second-least (0.94, [0.93, 0.96]), striatum most (0.83, [0.76, 0.89]). A permutation null model (500 shuffles of trial order per area-session) confirmed the drift is not measurement noise: observed centroid similarity (0.924, [0.915, 0.934]) was far below the shuffled expectation (0.995, [0.995, 0.996]; mean $z = -44.7$, $[-49.2, -40.4]$). Drift accumulates gradually across the session rather than in discrete steps (early-to-mid 0.942, mid-to-late 0.941; paired $t = 0.30$, $p = 0.77$). Striatum encodes action-reward associations with reliable relational structure even as its baseline firing rates shift, the pattern expected from a region that updates value continuously but reads out those values consistently (McClelland et al., 1995). Hippocampus does the opposite: the mean population state is preserved while internal structure reorganizes, which fits a region that forms new memories rapidly rather than maintaining fixed codes.

2.2 Geometric stability predicts trial-by-trial neural-behavioral coupling; temporal measures do not

For each area-session, the Spearman correlation between neural state magnitude (L2 norm of the trial population vector) and trial outcome (correct vs. incorrect) gave a behavioral coupling score. Shesha predicted this ($\rho = 0.18$, 95% CI $[0.05, 0.31]$, $p = 0.005$, $n = 229$). Centroid drift did not ($\rho = 0.002$, $[-0.13, 0.13]$, $p = 0.976$). Neither did a whitened unbiased cosine metric (Diedrichsen et al., 2021) ($\rho = 0.089$, $[-0.04, 0.21]$, $p = 0.180$).

At the session level ($n = 26$), Shesha did not predict mean task accuracy ($\rho = 0.087$, $p = 0.191$) or accuracy change over the session ($\rho = -0.079$, $p = 0.701$). Behavioral relevance of representational geometry is trial-to-trial, not session-to-session.

2.3 Olfactory recordings motivate a circuit-level account of geometric stability

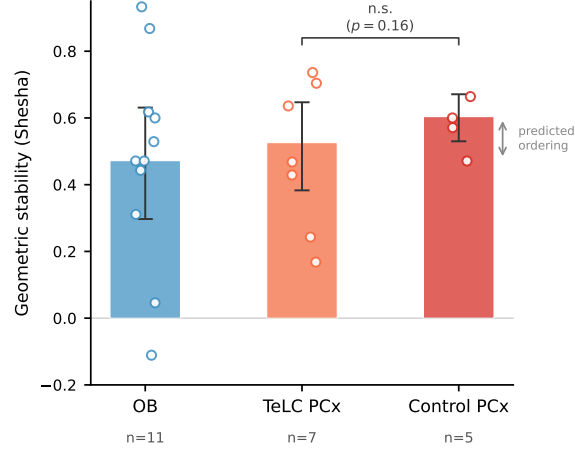


Figure 2: **Recurrent circuitry predicts geometric stability in the olfactory hierarchy.** Geometric stability (Shesha) in three groups from the Bolding and Franks (2018) PCX-1 dataset. OB: olfactory bulb recordings ($n = 11$); TeLC PCx: piriform cortex with recurrent connections silenced by tetanus toxin light chain ($n = 7$); Control PCx: contralateral intact piriform cortex ($n = 5$). Bars show mean with 95% bootstrap confidence intervals; open circles show individual recordings. The predicted ordering $OB < TeLC\ PCx < Control\ PCx$ is observed. Bracket: Wilcoxon signed-rank test, Control vs. TeLC, $p = 0.16$ (n.s.).

The Steinmetz results establish that geometric stability varies across brain regions and predicts behavior, but they do not explain why. Piriform cortex offers a natural test case: its feedforward input (from olfactory bulb) and recurrent circuitry are anatomically separable, and tetanus toxin light chain (TeLC) selectively eliminates recurrent excitatory connections (Bolding & Franks, 2018; Bolding et al., 2020). Simultaneous

OB and PCx recordings from the PCX-1 dataset (Bolding & Franks (2018); 11 paired sessions, 6 odors at 0.3% v/v) showed the predicted ordering: OB (0.47) < TeLC PCx (0.53) < Control PCx (0.60). The sample sizes are small ($n = 11$ paired, $n_C = 5$, $n_T = 7$) and neither comparison reaches significance (OB vs. PCx: $p = 0.35$; Control vs. TeLC: $p = 0.16$). We do not treat these as confirmatory results. Rather, the olfactory circuit provides the necessary biological specification for the attractor model that follows: a feedforward pathway with variable input, recurrent connections that could stabilize geometry, and a manipulation that removes them.

2.4 Recurrent coupling increases geometric stability: an attractor network account

We tested whether recurrent coupling stabilizes geometry in an E/I-balanced rate network ($N = 200$ units) receiving sparse feedforward input with 70% channel dropout per trial, simulating the incomplete OB-to-PCx projection (see Section 3 for full architecture and parameters). The logic follows directly from the formalism: if trial-to-trial dropout means each split receives a different random subset of the input, then the RDMs defined by Eq. 1 will differ across splits and S (Eq. 2) will be low. Recurrent dynamics (Eq. 3) counteract this by attracting responses to stimulus-specific fixed points regardless of which input channels survive. At $J = 0$ (no recurrence; TeLC analog), the network receives only 30% of the stimulus per trial and Shesha is low (0.27). As J increases, recurrent dynamics complete the pattern from partial input. At $J = 1.4$, Shesha recovers to 0.51. Across 10 independently seeded networks and 15 values of J , Shesha increased monotonically (Spearman $\rho = +0.64$, $p = 0.010$). Within-session RDM

consistency was less sensitive to J ($|\rho| = 0.55$), consistent with the idea that geometric and temporal measures respond differently to circuit parameters.

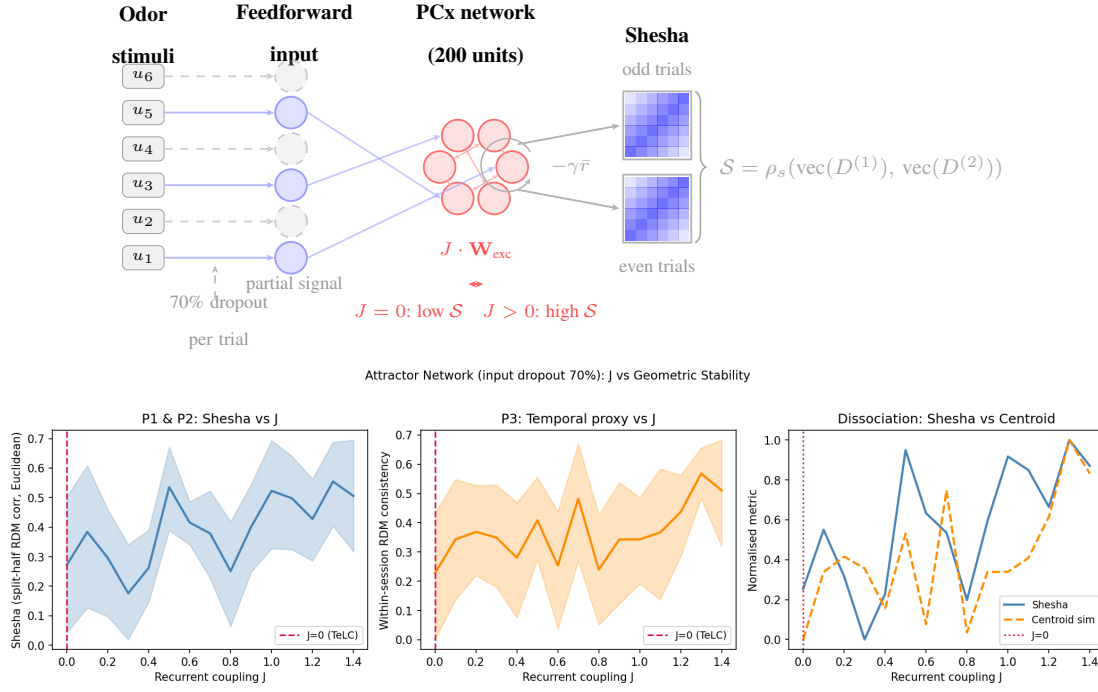


Figure 3: Recurrent coupling increases geometric stability via pattern completion in a rate network model.

Figure 3: **Figure 3 (continued): Recurrent coupling increases geometric stability via pattern completion in a rate network model.** An E/I-balanced rate network ($N = 200$ units) received sparse feedforward input with 70% channel dropout per trial, simulating the incomplete OB-to-PCx projection. Results are shown across 10 independently seeded networks and 15 values of recurrent coupling strength J (0 to 1.4); shaded regions show bootstrap 95% CIs. **(Top)** Model schematic. Six odor stimuli drive a 200-unit E/I-balanced rate network through sparse feedforward connections ($\mathbf{W}_{\text{ff}}$); 70% of input channels are randomly zeroed per trial (dashed arrows), simulating the incomplete OB-to-PCx projection. Recurrent excitatory connections scale with coupling strength J ; a fixed global inhibitory leak ($-\gamma\bar{r}$) maintains E/I balance independently of J . Geometric stability $\mathcal{S}$ is computed as the Spearman rank correlation between RDMs from odd and even trial subsets. At $J = 0$ (TeLC analog), incomplete feedforward drive produces inconsistent trial responses and low $\mathcal{S}$; increasing J engages attractor dynamics that complete the pattern, raising $\mathcal{S}$. **(Bottom, Left)** Shesha increases monotonically with J (Spearman $\rho = +0.64$, $p = 0.010$). At $J = 0$ (TeLC analog, dashed red line), Shesha = 0.27; at $J = 1.4$, Shesha = 0.51. **(Bottom, Center)** Within-session RDM consistency (temporal proxy) as a function of J . **(Bottom, Right)** Normalized comparison: Shesha is more sensitive to J ($|\rho| = 0.64$) than within-session consistency ($|\rho| = 0.55$), consistent with the geometric-temporal dissociation observed in the Steinmetz data.

3 Materials and Methods

3.1 Steinmetz dataset

Neuropixels recordings from Steinmetz et al. (2019): 26 sessions (≥ 60 trials), 229 area-sessions (≥ 10 neurons), 68 brain areas, visual contrast discrimination task. Stimulus conditions: 9 contrast pairings. Decision epoch: 0–500 ms post-stimulus; spike counts averaged over epoch. Each trial population vector was L2-normalized before computing RDMs. Brain areas grouped into 8 functional regions: Visual, Thalamus, Motor, Frontal, Hippocampus, Striatum, Midbrain, Other.

3.2 PCX-1 dataset

Silicon probe recordings from the PCX-1 dataset (Bolding & Franks (2018); CRCNS.org, doi:10.6080/K00C4SZB): 32-channel NeuroNexus Poly3 probes, Spyking-Circus spike sorting. Awake trials selected via the FT:LT series indices from ExperimentCatalog files. Population vectors: first-sniff spike rates (`MultiCycleSpikeRate`, sniff index 0) per unit per odor presentation. Units with mean rate below 5% of the recording’s median positive rate were excluded. Vectors were square-root transformed and L2-normalized. Two recordings were excluded as outliers: session 170621 (hardware failure: 5 of 6 odor valves returned NaN) and session 150221 bank 2 (mean firing rate 2.58 Hz, an order of magnitude below the dataset median). Simul experiment: 11 paired OB and PCx recordings (loading configurations A and C; 6 odors at 0.3% v/v; ExperimentCatalog valve indices per Table 1 of the data description). TeLC-PCx experiment: 5 control and 7 TeLC recordings after exclusions.

3.3 Geometric stability (Shesha)

Trials were split into odd and even series. Condition-averaged population vectors were computed for each odor condition in each half. The RDM was the matrix of pairwise cosine distances between condition centroids. Shesha is the Spearman correlation of the upper triangular vectors of the two RDMs (Raju, 2026a,b). Bootstrap 95% CIs: 10,000 resamples.

3.4 Temporal stability and behavioral coupling

Centroid drift: each session split at the median trial; cosine similarity of early and late normalized population centroids. Permutation null: 500 shuffles of trial order per area-session. Neural-behavioral coupling: Spearman correlation between the L2 norm of each trial’s population vector and trial outcome (correct/incorrect). This per-area-session value was then correlated across area-sessions with each stability metric using Spearman ρ .

3.5 Rate network model

PCx principal neurons receive input from only a sparse, random subset of olfactory bulb projections on any given trial (Bolding & Franks, 2018). Recurrence could stabilize geometry by completing the full stimulus pattern from these partial inputs, an attractor mechanism (Hopfield, 1982; Haberly, 2001). We tested this in a rate network model with the following architecture.

Network. $N = 200$ rate units with 20% sparse random recurrent connectivity. The recurrent weight matrix $\mathbf{W}_{\text{rec}}$ was drawn from a Gaussian and normalized by $\sqrt{k}$, where

k is the mean in-degree, then split into non-negative excitatory ($\mathbf{W}_{\text{exc}}$) and non-positive inhibitory ($\mathbf{W}_{\text{inh}}$) parts. The dynamics are central to the stability argument:

$$\tau \dot{x} = -x + J\mathbf{W}_{\text{exc}}f(x) + \mathbf{W}_{\text{inh}}f(x) - \gamma\bar{r} + \mathbf{W}_{\text{ff}}u + \eta, \quad (3)$$

where $f(x) = \max(x, 0)$, $\tau = 20$ ms, $\Delta t = 1$ ms, $\gamma = 0.4$ (global inhibitory leak), and η is Gaussian noise ($\sigma = 0.05$, scaled by $\sqrt{\Delta t/\tau}$). The parameter J in Eq. 3 controls the strength of recurrent excitation and is the single free parameter we sweep. At $J = 0$, the excitatory recurrent term vanishes and the network is a purely feedforward relay: each trial’s response is determined entirely by which input channels survive dropout, so trial-to-trial variability in the RDM is high and $\mathcal{S}$ (Eq. 2) is low. As J increases, the recurrent term $J\mathbf{W}_{\text{exc}}f(x)$ pulls activity toward stimulus-specific fixed points, completing the full pattern from partial input. This is the mathematical mechanism by which recurrence increases geometric stability: it reduces the dependence of the population response on the stochastic input mask, making the RDM reproducible across trial splits. Only $\mathbf{W}_{\text{exc}}$ was scaled by J ; inhibition was held fixed to maintain E/I balance across the J sweep. All networks operated in a stable fixed-point regime throughout the J range tested: activity was clipped at ± 10 to prevent divergence and no more than 1% of values reached the boundary at any $J \leq 1.4$.

Inputs. Nine stimulus conditions were represented as fixed 50-dimensional random unit vectors u_c , drawn once per network seed and held constant. The feedforward weight matrix $\mathbf{W}_{\text{ff}} \in \mathbb{R}^{N \times 50}$ was drawn from $\mathcal{N}(0, 1/\sqrt{50})$. To simulate the incomplete and variable projection from olfactory bulb to piriform cortex (Bolding & Franks, 2018), 70% of the 50 input channels were independently zeroed on each trial (input dropout). Each trial therefore receives a different random 30% of the full stimulus,

mimicking the sparse combinatorial nature of the OB-to-PCx projection. At $J = 0$ (no recurrence; the TeLC analog), the network cannot recover the full stimulus pattern from 30% of the input: responses vary across trials and Shesha is low (0.27). As J increases, recurrent dynamics pull activity toward stimulus-specific fixed points, completing the pattern from partial input. At $J = 1.4$, Shesha recovers to 0.51.

Protocol. Each network was initialized with small random activity and settled for 400 ms before a 150 ms collection window; the time-averaged firing rate over the collection window gave the population vector for that trial. 20 noise trials per condition were collected per network. Geometric stability was computed from raw (non-normalized) firing rates using Euclidean-distance RDMs, since L2-normalization would collapse vectors onto the unit sphere and hide the magnitude amplification from recurrence. 10 independently seeded networks were run at each of 15 values of J uniformly spaced from 0 to 1.4. Random seed 320 was used throughout.

Results. Shesha increased monotonically with J (Spearman $\rho = +0.64$, $p = 0.010$, $n = 10 \text{ networks} \times 15 \text{ coupling values}$). Within-session RDM consistency (a temporal proxy computed from early vs. late trial halves) was less sensitive to J ($|\rho| = 0.55$ vs. 0.64 for Shesha), consistent with the geometric-temporal dissociation observed in the Steinmetz data. We note that the model’s temporal proxy measures within-session sampling noise rather than longitudinal drift, so the parallel to the Steinmetz result is qualitative.

3.6 Statistics

All correlations: Spearman ρ with bootstrap 95% CIs (10,000 resamples). Paired comparisons (OB vs. PCx; Control vs. TeLC): one-sided Wilcoxon signed-rank test in the direction of the prediction. Random seed 320 was used throughout.

4 Discussion

The Steinmetz data reveal something counterintuitive. Striatum drifts the most of any region measured (centroid similarity 0.83) yet has the most stable representational geometry (Shesha 0.44). Hippocampus drifts the least (0.94) yet has the least stable geometry (0.19). These two measures are not redundant: they are close to orthogonal across the regional hierarchy.

This dissociation is the central empirical finding, and it is not a consequence of using Shesha specifically. It is a fact about the data: the ranking of brain regions by within-session geometric reliability is approximately the reverse of their ranking by centroid preservation. No prior analysis of the Steinmetz dataset reported this inversion, because no prior analysis asked whether within-session pairwise geometry is reliable independently of whether centroids are preserved. The finding is visible in raw split-half RDM correlations before any methodological choices specific to Shesha are made.

One might ask whether decoding accuracy captures the same information more directly. It does not, and the data show this. Mean decoding accuracy across area-sessions is uncorrelated with Shesha ($\rho = 0.09$, $p = 0.19$, $n = 228$) and uncorrelated with trial-by-trial neural-behavioral coupling ($\rho = 0.01$, $p = 0.88$). Shesha predicts behavioral

coupling ($\rho = 0.18$, $p = 0.005$); decoding does not. Decoding accuracy measures whether task variables are linearly separable in a population—a question about information content. Shesha measures whether the pairwise distance structure among conditions reproduces across independent trial subsets—a question about representational reliability. A population can be highly decodable yet geometrically brittle if information is concentrated in a single low-dimensional subspace. The behavioral coupling result shows that this distinction matters: what predicts how tightly a region’s activity tracks behavior on individual trials is not whether it decodes well, but whether its geometry is reliable.

An obvious objection is that Shesha tells us nothing about drift since it does not consider temporal evolution. This is true by design, and the distinction is the point. Centroid drift is the dominant framework for understanding behavioral stability during representational reorganization (Gallego et al., 2020; Rule et al., 2019). Our data directly test whether centroid preservation predicts behavioral coupling and find that it does not ($\rho = 0.002$, $p = 0.976$). Shesha is not a drift measure; it is a reliability measure that operates within a session. The fact that it predicts behavior where drift does not suggests that within-session geometric reliability is a distinct and functionally relevant property, not a restatement of temporal stability in different language.

This is not an isolated observation. Keinath et al. (2022) imaged CA1 over a month and found drift unfolds orthogonally to the context representation: cells reorganize, but the population geometry for context is preserved. Schoonover et al. (2021) documented substantial drift in piriform cortex over weeks while odor identity remained decodable. Deitch et al. (2021) reported the same pattern in visual cortex. The consistent picture

across regions and species is that drift at the neuron level does not necessarily degrade geometry at the population level. Shesha quantifies the within-session version of this: whether the pairwise distance structure among conditions is consistent across independent trial samples, on a timescale of minutes rather than weeks. It is separable from centroid drift by construction, and it predicts behavioral coupling where drift does not.

Aitken et al. (2022) proposed that drift has geometric structure, preferentially affecting task-null dimensions while preserving task-relevant geometry (Driscoll et al., 2022). Shesha is complementary rather than redundant. A region can have geometrically structured drift in the Aitken et al. sense and still have low Shesha if within-session noise is large. The two measures operate at different timescales and catch different failure modes.

The olfactory data are not a second empirical finding; they are the biological grounding that necessitates the attractor model. Neither the OB vs. PCx comparison ($p = 0.35$, $n = 11$) nor the TeLC manipulation ($p = 0.16$, $n_C = 5$, $n_T = 7$) reaches conventional significance thresholds. Post-hoc power for the TeLC comparison at the observed effect size ($d \approx 0.5$) is approximately 20%, and the minimum achievable p -value with $n_C = 5$ in a one-sided Wilcoxon test is $1/2^5 = 0.031$. What the olfactory data provide is not statistical confirmation but a specific circuit prediction: if recurrent connections stabilize representational geometry, then removing them (TeLC) should pull piriform stability toward olfactory bulb levels, and the three-group ordering OB (0.47) < TeLC PCx (0.53) < Control PCx (0.60) should hold. The data are directionally consistent with this prediction. The rate network model (Eq. 3) then formalizes the mechanism: recurrent coupling amplifies split-half RDM consistency by completing stimulus patterns

from sparse feedforward input ($\rho = +0.64$, $p = 0.010$). The model is the argument; the olfactory data are its biological motivation. Piriform cortex has long been modeled as an auto-associative network (Haberly, 2001; Hopfield, 1982), and Bolding et al. (2020) showed directly that recurrent connections stabilize odor representations across brain states. The TeLC result here extends that to geometric stability specifically. A reversible manipulation with a larger sample would be needed to confirm the effect.

A recent paper using the same Bolding-Franks dataset (Morales et al., 2025) attributes piriform drift to slow synaptic weight fluctuations, with fast learning compressing representations onto a lower-dimensional manifold. Lower dimensionality is consistent with higher Shesha: a more structured representation should produce more consistent RDMs across trial splits. Their result and ours concern different properties of the same circuit and are not competing claims. Wagner et al (Wagner et al., 2026) provide a complementary theoretical perspective. Their recurrent networks trained on hippocampal prediction tasks converge to a mixed-symmetry regime in which dominant symmetric recurrence stabilizes an attractor manifold while a weaker antisymmetric component drives sequential flow. This is the same computational logic underlying our rate network model (Eq. 3): symmetric recurrent coupling (JW_{exc}) stabilizes stimulus-specific fixed points and increases geometric stability, while the network’s capacity for pattern completion from partial input is what makes the RDM reproducible across trial splits. Their finding that this regime is task-selected rather than architecturally imposed strengthens the case that recurrent stabilization of geometry is a general principle of cortical computation, not an idiosyncrasy of piriform cortex. The model predicts that geometric stability should scale with recurrent coupling strength, and this prediction

is confirmed in the controlled olfactory circuit where sensory input is held constant. Across the whole-brain Steinmetz hierarchy, however, the recurrence gradient does not predict Shesha ($\rho = -0.30$, $p = 0.27$, $n = 15$ non-hippocampal areas, using a curated within-area recurrence index compiled from published anatomy). This suggests that other factors, particularly the strength and reliability of sensory drive, dominate the between-area variation in geometric stability when circuit architecture is not controlled. The olfactory comparison isolates recurrence by holding input constant; the whole-brain data do not.

Returning from circuit mechanism to whole-brain function: the behavioral coupling result ($\rho = 0.18$, $p = 0.005$, $n = 229$) is modest in magnitude but shows that Shesha is tracking something functionally real, even if coarsely. A region with unstable pairwise geometry will produce unreliable input to downstream areas regardless of what those areas compute, constraining the fidelity of information transmitted through inter-area communication subspaces (Weiss et al., 2025). The correlation connects to a broader question about whether geometric properties of neural manifolds predict downstream computation (Li et al., 2024). Shesha is a coarse measure of that reliability: it captures the full RDM rather than task-specific subspaces, but the behavioral correlation shows it is tracking something real. Geva et al. (Geva et al., 2023) showed that time and experience have separable effects on hippocampal drift, consistent with the idea that temporal and geometric stability reflect distinct circuit properties (Natrajan & Fitzgerald, 2025).

Several limitations apply. The OB-PCx and TeLC comparisons come from separate experiments. Shesha as computed here is a global metric: it does not localize instability to specific stimulus pairs or subspaces, and given that drift is geometrically struc-

tured (Aitken et al., 2022) there is reason to think geometric stability may be as well. Finally, the two datasets differ on most dimensions that matter: species, task, recording technology, number of conditions. The Steinmetz result is about visual decision-making across the whole brain; the PCX-1 result is about olfactory processing in one cortical area. Connecting them through a shared principle, that recurrent circuitry stabilizes representational geometry, is our interpretation, not a direct claim of the data.

The same metric applied to artificial neural networks shows that geometric stability and transfer performance are dissociable (Raju, 2026a): models that perform best can have the worst stability, and vice versa. Whether this reflects a principle common to biological and artificial systems is not settled by the present results, but the parallel is suggestive.

Declarations

Conflict of interest The authors declare no competing interests.

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the authors, and the authors assume full responsibility for all content and claims in this work.

Code and Data Availability

All code needed to reproduce the analyses and the computational models is publicly available at <https://github.com/prashantcraju/neuroscience-drift>.

The geometric stability metric is implemented in the `shesha-geometry` Python package, available on PyPI (`pip install shesha-geometry`; Raju (2026b)).

The two datasets used in the study (Steinmetz et al., 2019 and Bolding & Franks, 2018) are both publicly available. The GitHub repository contains code for automatically downloading the Steinmetz data and instructions for downloading the Bolding and Franks data.

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